Session Summary — 2026-05-08 CRM Wave 1 + Cost Strategy

Date: 2026-05-08 **Branch**: `feature/openai-realtime-migration` **Sprint**: CRM Wave 1 — Phone-Keyed Returning Customer + Email NATO fix + silent agent diagnosis/fix + 3-PDF cost strategy **Pilot Position**: M1.4 / M9 (25% — single-store live verification)

🎯 Achievements (12 commits)

A. CRM Wave 1 — Spec MUST 100% complete + live verified

Commit	Scope	LOC
`ae04d8e`	spec design (582 lines)	docs
`682ecab`	C1 — DB migration (5 NULLABLE columns + index)	+64
`c0282fa`	C2 — services/crm/ + update_call_metrics	+372
`6090cdb`	C3 — customer_context block (INVARIANTS-prior)	+130/-1
`c674643`	C4 — realtime_voice lookup hook + AHT call_end	+103/-2
`85f9b3f`	C5 — 66 unit tests Tier 1-3	+689

B. Live verification (4 calls, 124s avg)

- ✅ **Welcome back, Michael!** (turn 1, visits=27)
- ✅ **Email auto-fill** "I have cymeet@gmail.com on file"
- ✅ **"the usual?" 발동** `[crm] usual_offered=true detected` (10:27 call)
- ✅ DB persist 3 rows (`crm_returning=True` , `crm_visit_count=26/1/27` , `call_duration_ms=115279/186216/102120`)
- ✅ AHT 절감 추정: returning calls ~13s saved via email auto-fill

C. Bonus follow-up fixes

Commit	Issue	Fix
`9f09604`	Email NATO `cym eet@gmail.com` 공백 (Call #2)	last_email_recital_text 별도 latch, last_assistant_text 덮어쓰기 방지
`2b26f2a`	Silent agent 진단 부재	response.done 핸들러에 id/status/output_types/transcript/status_details 캡처
`f6a73f6`	Silent agent root cause = TPM 40K rate_limit_exceeded	retry-after 파싱 + asyncio.sleep + response.create 자동 재시도 (max 3/call)

D. Cost Strategy PDFs (3 reports)

Commit	PDF	Pages	Topic
`40e098f`	openai-realtime-cost-strategy.pdf	17	Q1-4 통합 + per-call/min cost
`6dd2640`	realtime-models-3way-comparison.pdf	15	1.5 vs 2.0 vs mini + JM 적합도 0-50
`294e607`	hybrid-rollout-timing-and-risk.pdf	13	Pilot 단계 진단 + Hybrid 최적 시점
`8390f62`	(cost-strategy 갱신)	+5p	Hybrid 4-시나리오 비교 추가

📊 Live Diagnostic Capture — Silent Agent Root Cause

라이브 데이터 (2026-05-08 10:28:43):

```
status=failed type=failed
error.code=rate_limit_exceeded
error.message='Rate limit reached for gpt-realtime-1.5
  (for limit gpt-4o-realtime). Limit 40000, Used 30058,
  Requested 10244. Please try again in 1.903s'
```

진단 결과: 우리가 봤던 모든 silent-agent 사례 = **OpenAI TPM 40K (Tier 1) 한계**.

- Call #1 (09:38): turn 7 후 11s 무음 — TPM 한계 추정 (진단 없어 못 봄)
- Call #4 (10:27): turn 8 explicit fail with retry_after=1.903s
- 사용자 체감 무음 6-10초 = retry-after (1.9s) + caller 반응 시간 (5s)

Fix 전략 (2단): 1. **외부**: Tier 2 도달 ($50 누적 + 7일) → TPM 40K → 200K (5×) 2. **내부**: `f6a73f6` retry 코드 — silent 무음 6-10s → 1-2s 자동 회복

📐 Per-Call Cost Reconciliation (실측)

평균 통화 124s / 12.9 turns / 6,500-token system prompt:

Stream	Tokens	gpt-realtime-1.5 cost
Audio in	833	$0.027
Audio out	1,860	$0.119
Text in (fresh)	32,550	$0.130
Text in (cached, 80% hit)	86,400	$0.035
Text out	1,040	$0.017
OpenAI subtotal		**$0.327**
Twilio (2.07 min)		$0.037
All-in per call		**$0.364**
Per minute		**$0.176**

3-Way 비교: | Model | Per call | Per min | 5-store/mo | 500-store/mo | |---|---|---|---|---| | gpt-realtime-1.5 (current) | $0.405 | $0.196 | $1,823 | $182,250 | | gpt-realtime-2.0 (new) | $0.408 | $0.197 | $1,836 | $183,600 | | gpt-realtime-mini | $0.150 | $0.072 | $675 | $67,500 | | **Hybrid Routing** ⭐ | **$0.284** | **$0.137** | **$1,278** | **$127,800** |

🛡️ Protected Code Zones (Updated)

이번 세션에서 production-ready 확정된 9개 추가:

1. `services/crm/customer_lookup.py` — phone lookup, 500ms timeout, graceful degrade
2. `services/bridge/transactions.update_call_metrics` — background-safe PATCH
3. `voice_websocket.py:_render_customer_context_block` — INVARIANTS 직전 배치
4. `realtime_voice.py customer_lookup hook` — env var CRM_LOOKUP_ENABLED 가드
5. `realtime_voice.py session_state` 7개 신규 키
6. `realtime_voice.py response.done` 진단 로그 (id/status/output_types/transcript)
7. `realtime_voice.py rate_limit_exceeded retry` — silent 자동 회복
8. `bridge_transactions` 5 신규 컬럼
9. `idx_bridge_tx_phone_store` partial index

📋 Pilot Roadmap Status

```
M0 ✅ Wave A.3 hardening
M1.1 ✅ Wave A.3 라이브 검증 (오늘 09:30)
M1.2 ✅ CRM Wave 1 라이브 검증 (오늘 10:00)
M1.3 ✅ Silent agent 진단 (오늘 11:00)
M1.4 ✅ Retry fix 적용 (오늘 EOD)            ← 세션 종료 시점
M1.5 ⏸ Tier 2 도달 (사용자 $50 충전 + 7일)
M1.6 ⏸ 1매장 24-48h 무인 운영
M2   ⏸ KPI baseline 보고서
M3   ⏸ 5매장 deployment
M4   ⏸ 5매장 안정 운영
M5   ⏸ KPI + Series A pre-prep
M6   ⏸ A/B test 인프라 (Step A read-only)
M7   ⏸ 2.0 canary (Step B)
M8   ⏸ ⭐ Hybrid Routing (Step C-E)
M9   ⏸ Vertical 확장 (KBBQ/Sushi)
```

🔔 Outstanding User Actions

즉시 (이번 주)

1. **OpenAI Dashboard**: $50 충전 (Buy credits)
2. **Tier 2 자동 승급** 7일 대기

1-2주 내

3. **Live 검증 통화** (Tier 2 도달 후) — silent agent 사라졌는지 확인

4. KPI baseline 측정 (M2)

4-6주 내

5. 5매장 Twilio 번호 추가 + store 라우팅 (M3)
6. KPI 보고서 + Series A pre-prep deck (M5)

🚨 Outstanding Issues (Backlog)

#	Issue	Priority	후속 sprint
1	Welcome back not fired when name=None (visit_count≥1, recent[]=empty)	★	Wave 2
2	Modifier 정확도 (large→medium STT)	★★	별도 sprint
3	"forget me" voice command (Cat 6.3)	★	Wave 2
4	Tiered opt-in (T2 personalization)	★	Wave 2
5	customers 테이블 도입 (denormalized → 정규화)	★	Wave 2
6	Mini KO/JA/ZH 50-utterance eval	★★	Step C 직전 (M7)
7	gpt-realtime-2.0 canary 검증	★★	Step B (M7)

📚 Key Documents Created This Session

- `docs/superpowers/specs/2026-05-08-crm-wave-1-design.md` — Spec
- `docs/cost-analysis/2026-05-08/openai-realtime-cost-strategy.{md,html,pdf}` — 종합 비용 전략
- `docs/cost-analysis/2026-05-08/realtime-models-3way-comparison.{md,html,pdf}` — 3-way 모델 비교
- `docs/cost-analysis/2026-05-08/hybrid-rollout-timing-and-risk.{md,html,pdf}` — Hybrid 시점·위험
- `docs/cost-analysis/2026-05-08/_build_pdf.py` — PDF 빌드 스크립트 (재사용)
- 본 세션 요약 (이 파일)

🔧 Memory Updates

- ⭐ 신규: `session_resume_2026-05-08_crm-wave-1.md` (다음 세션 진입점)
- 신규: `feedback_implementation_first_testing.md` (TDD override)
- 갱신: `feedback_db_schema_reference.md` (도메인 한정)
- 갱신: MEMORY.md 인덱스 (위 신규/⭐ 추가)

🏁 Next Session Entry

```
# 다음 세션 첫 명령어
# 1. session_resume_2026-05-08_crm-wave-1.md 읽기
# 2. 환경 점검
git log --oneline -15
curl -s -o /dev/null -w "local=%{http_code}\n" http://127.0.0.1:8000/health
curl -s -o /dev/null -w "ngrok=%{http_code}\n" https://bipectinate-cheerily-akilah.ngrok-free.d

# 3. OpenAI Tier 2 도달 확인 (Settings → Limits)
# 4. 사용자 결정 옵션:
#    α) Tier 2 검증 통화 (silent agent 사라졌는지 라이브 확인)
#    β) M2 1매장 24-48h baseline KPI 수집
#    γ) M3 5매장 deployment 계획
#    δ) Modifier 정확도 sprint (별개)
```

End of Session Summary.

Distribution: founder + technical handoff.